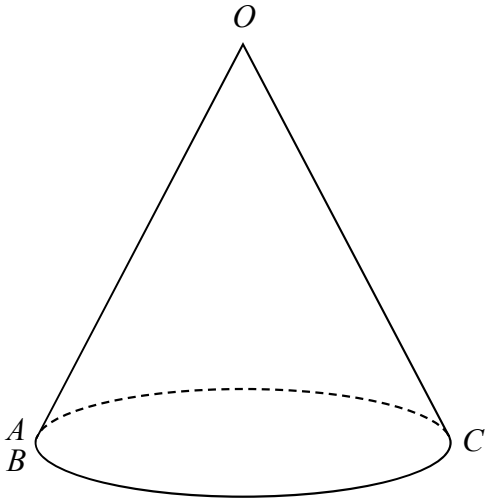
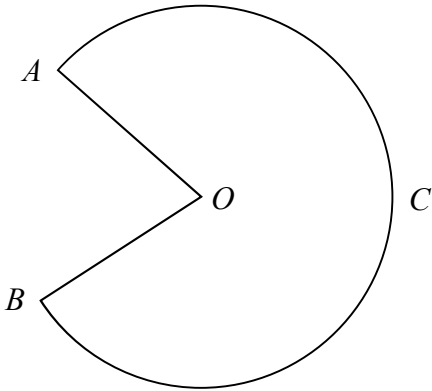


23 The diagram shows a sector $OACB$ of a circle with centre O .
 The point C is the midpoint of the arc AB .

The diagram also shows a hollow cone with vertex O .
 The cone is formed by joining OA and OB .

Volume of cone = $\frac{1}{3} \pi r^2 h$
 Curved surface area of cone = $\pi r l$



The cone has volume 56.8 cm^3 and height 3.6 cm .

Calculate the size of angle AOB of sector $OACB$.
 Give your answer correct to 3 significant figures.
 You must show all your working.

